

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting Large Language Models (LLMs) underpins provenance verification and model attribution, yet the operating regime of existing methods is markedly narrower than the problem demands. Prior fingerprinting techniques are validated almost exclusively after fine-tuning, once a model has already accumulated stable signatures from its training data, optimization trajectory, and hyperparameters. Because the bulk of a model’s capacity and knowledge is acquired during pre-training rather than downstream adaptation, large-scale pretraining is the more fundamental regime for lineage verification—and it is precisely this regime in which we find existing methods to be *unreliable*: their signatures materialize only after substantial training, so they operate as post-hoc identifiers rather than as the intrinsic, lifelong marks that the classical Galtonian conception of a fingerprint demands. We therefore formulate a stronger and genuinely intrinsic notion of LLM fingerprinting and realize it in **SeedPrints**, which reads out the prediction biases induced by random initialization as persistent, seed-dependent identifiers—signals that are fixed before a single gradient step is taken and are recovered from white-box access to a model’s outputs (its logits or final-layer hidden states). We show that untrained models already exhibit reproducible, seed-specific prediction biases, and that these weak signals remain statistically detectable throughout training, supporting high-confidence verification of whether a model shares initialization lineage with a candidate reference. Whereas prior techniques fail during early pre-training or degrade under distribution shift, **SeedPrints** remains effective across the entire training lifecycle, from random initialization through large-scale pretraining to downstream adaptation. On from-scratch LLaMA-style and Qwen-style models it establishes seed-level distinguishability and birth-to-lifecycle identity verification, while evaluations on OLMo-2-7B pretraining trajectories and a real-world benchmark of open-weight models confirm its robustness under prolonged training, pronounced domain shift, and diverse parameter-altering transformations. Together, these results establish that initialization imprints a persistent, seed-specific identity on an LLM—an intrinsic fingerprint present at “birth” and legible across the model’s lifecycle. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprinting has emerged as a principled means of identifying, attributing, and tracing large language models through their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). Its central objective is to furnish model owners with a verifiable link between a suspicious model and its putative original, thereby exposing model theft and unauthorized reuse (Yoon et al., 2025; Zhang, 2025). As proprietary models are increasingly released, adapted, and redistributed, the trustworthiness of this link—sustained across the full span of a model’s development rather than only its final deployed form—is what determines whether attribution can be relied upon in practice.

Much of this literature explicitly inherits the metaphor of biological fingerprints from Francis Galton’s *Finger Prints* (1892) (Galton, 1892):

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

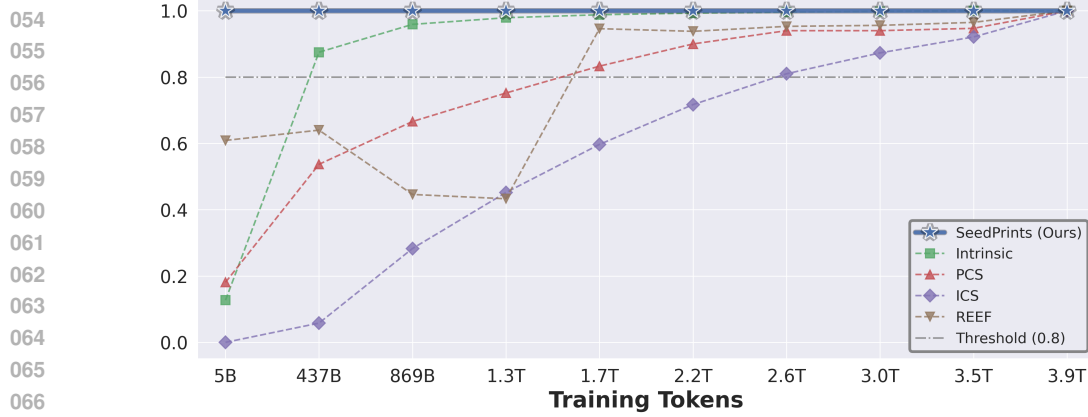


Figure 1: **Existing fingerprinting methods fail to detect model lineage during early pretraining, whereas SeedPrints succeeds from the first checkpoint.** We compare five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint as the reference model. The y -axis reports the lineage score (higher indicates a stronger lineage signal), and the dashed line marks the 0.8 detection threshold. Every baseline degrades below this threshold at early checkpoints, failing precisely where verification matters most; **SeedPrints**, by contrast, detects lineage with overwhelming significance ($p \ll 0.001$, plotted as $1 - p$) from the very first checkpoint and strengthens monotonically thereafter.

*“A fingerprint is the pattern formed by friction-ridge skin on the fingertips;
this ridge configuration is individually unique and essentially permanent across an
individual’s lifetime.”*

This analogy imposes two precise and demanding criteria: an effective fingerprint must be both *unique*, distinguishing a model from independently produced counterparts, and *persistent*, present from the moment of a model’s “birth” at initialization and stable thereafter. Judged against this standard, prevailing fingerprinting approaches fall short (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As Figure 1 makes explicit, these methods do not reliably detect lineage during early pretraining—the very stage at which verification is most needed. The reason is structural: their signatures are defined only after a model has largely converged, being read from settled parameters or from generated text. The separability they exhibit therefore encodes not an intrinsic mark of the model but an imprint of surface-level training components—data signatures, optimization dynamics, and hyperparameters. This dependence has a concrete cost: as we show in Section 5.1, existing baselines are readily misled and fail to recover the correct lineage once the training data distribution shifts substantially. Such methods thus function as *post hoc* identifiers rather than as the innate, stable marks—Galtonian fingerprints—that accompany a model from its inception.

In this work, we advance a stricter and more principled notion of LLM fingerprinting: an intrinsic property fixed at *initialization* and recoverable at *any* subsequent point in training. Concretely, given white-box access to two models’ outputs, our method decides whether they share an initialization lineage—relating a suspect model to a candidate reference rather than recovering a numeric seed in the abstract. We show that this notion is not merely aspirational but empirically realizable, and we instantiate it in a method that verifies lineage from the moment of a model’s birth through large-scale pretraining and deployment. Our key contributions are as follows:

- **Exposing a structural limitation of existing fingerprints.** We establish that prior fingerprinting approaches fail to reliably detect lineage during early pretraining and are readily misled by substantial shifts in the training data distribution, demonstrating that their apparent reliability rests on late-stage, data-dependent signals rather than on intrinsic identity. (Section 5.1)
- **Identifying and operationalizing initialization-born fingerprints.** We show that untrained models already carry seed-dependent biases in their outputs under random inputs, constituting a weak but persistent identity signal fixed at initialization. (Section 3) Building directly on this observation, we introduce **SeedPrints**, which isolates these initialization-born signals on a shared subset of disfavored output dimensions and adjudicates lineage through a calibrated statistical test, rendering the decision robust to confounds such as data distribution, training duration, and optimization dynamics. (Section 4)

- **Empirical validation of seed-level identity.** We demonstrate that SeedPrints separates models that differ *only* in initialization seed—even under identical training pipelines and data orders—and that this identity persists under continual training across markedly divergent corpora, confirming that the signal it captures is neither a data proxy nor a training artifact. (Section 5.1)
- **Reliable verification across the full model lifecycle.** SeedPrints verifies lineage throughout large-scale pretraining trajectories (OLMo-2-7B Stage 1) without the early-stage failure that afflicts every baseline, and it extends to large-scale fine-tuning (up to 700B tokens) and to realistic deployment on the *LeaFBench* benchmark (Shao et al., 2025), which spans 65 open-weight models across 7 mainstream model families and 6 parameter-altering transformations (instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation). In these post-training settings SeedPrints matches the strongest dedicated baseline at near-perfect accuracy, while decisively outperforming all methods in the early-pretraining regime where they break down. (Sections 5.2 and 5.3)

2 RELATED WORK

Techniques for protecting LLM ownership fall into two complementary families: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a model’s pre-existing behaviors without modifying it (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025). Our work belongs to the passive family, yet departs from it in a decisive respect: it locates the signature at initialization rather than after convergence, and consequently supports verification throughout training rather than only at its endpoint.

Watermarking and Active Fingerprinting. Active approaches deliberately implant verifiable identifiers to support subsequent ownership checks. Classical instances include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018) and digital signatures or hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two forms predominate: *text watermarks*, which bias generation or embed predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.), and *model-weight watermarks*, which encode identifiers directly into parameters or bind them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are conceptually simple and can survive moderate fine-tuning (e.g., Dasgupta et al. 2024), their invasive nature presupposes control of the training process, which renders them unsuitable for retroactively marking third-party models—exactly the setting in which provenance disputes typically arise.

Passive LLM Fingerprinting. Passive fingerprinting instead identifies a model by analyzing its intrinsic properties without any modification, and the available techniques are organized primarily by the level of model access they assume. Under **white-box access**, signatures are extracted from model weights or internal computation, exploiting properties such as the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable direction of parameter vectors. Under **black-box access**, fingerprinting analyzes input–output behavior instead, using crafted queries (Pasquini et al., 2024), unique prompt–response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024), though such behavioral signatures tend to be less robust to fine-tuning. A single limitation cuts across both settings: these methods define their signatures *post hoc*, reading off properties that emerge only from a completed training process rather than the intrinsic marks present from random initialization that our work sets out to recover. Consequently, *they are not designed for lineage verification during large-scale pretraining*, where the post-training signals they depend on have not yet emerged. SeedPrints, operating in the same white-box regime as weight- and representation-based methods, is constructed expressly to close this gap.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

This section establishes the empirical foundation of our method: language models exhibit pronounced prediction biases that originate in the random initialization seed and remain detectable even after extensive training. These two properties—seed-specificity at initialization and persistence through optimization—instantiate the Galtonian requirements of uniqueness and permanence, and together they license the fingerprint developed in the sections that follow.

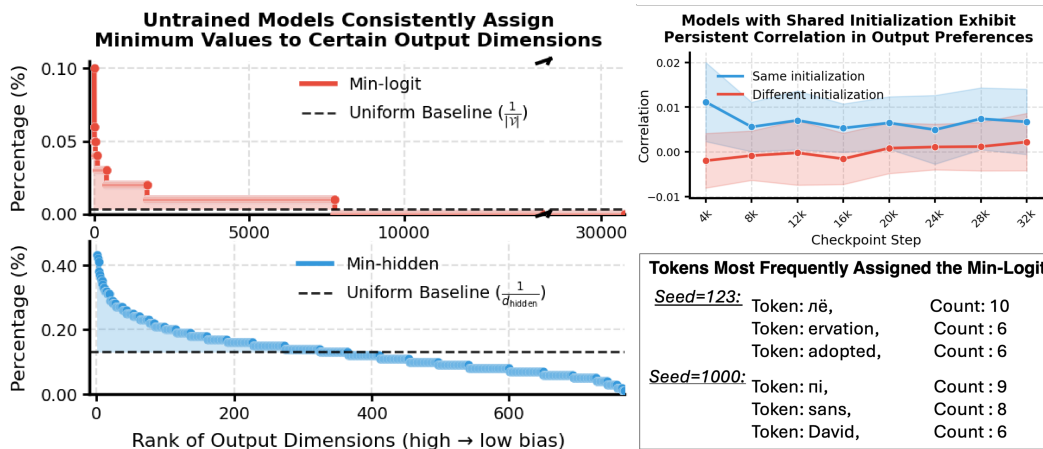


Figure 2: **Initialization-born output bias is seed-specific and persists through training.** *Left:* Under completely random inputs, the outputs of a randomly initialized LLaMA-2-style model are markedly non-uniform: the model systematically disfavors certain dimensions, which attain the minimum value far more often than chance. This bias is evident in both the logits (top, red) and the final hidden representations (bottom, blue); the dashed line marks the frequency expected under a uniform distribution, and the arrows in the top panel denote a broken x -axis that omits low-frequency tail ranks. *Upper Right:* Throughout training, a model retains a weak but consistent correlation in output bias with the initialization from which it descends. This correlation distribution is shifted upward relative to correlations against independently initialized models, evidencing that the initialization-born signal remains measurable rather than decaying away. *Lower Right:* The specific dimensions carrying strong bias are fixed by the initialization seed, illustrated here through the tokens that most frequently attain the minimum logit.

Initialization induces seed-specific bias profiles. We evaluate a LLaMA-2-style model initialized with seed 123 on 10,000 random input sequences of length 1,024, tokens sampled uniformly from the vocabulary. Figure 2 (*Left*) plots the frequency with which each output token (upper, from the final logits) and each hidden dimension (lower, from the last-layer representations) attains the minimum value across the 10,000 random trials. The pronounced non-uniformity of both distributions reveals a systematic output-preference bias: under purely random inputs, the model consistently disfavors a specific, seed-determined subset of coordinates. Repeating the experiment across initialization seeds shows that, while the overall magnitude of this bias is stable, the particular argmin dimensions are seed-dependent (Figure 2, *Lower Right*)—establishing precisely the per-seed distinctiveness a fingerprint requires. This behavior is consistent with recent evidence that biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026): under this account, distinct initialization seeds induce distinct contraction directions, which in turn yield the distinct argmin patterns we observe across output dimensions.

Training preserves initialization-born bias. Although training substantially rescales output magnitudes, it does not erase the *relative* preference over output dimensions imprinted at initialization. We train the initialized model (seed 123) for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate its intermediate checkpoints. Restricting attention to the $m = 50$ output dimensions most frequently assigned the minimum value by the initialized model, we compute, for each checkpoint, the correlation between its responses and those of the initialized model across random inputs on exactly these dimensions; this quantifies the degree to which the checkpoint continues to rank and disfavor the same coordinates as its initialization. As a reference, we compute the identical statistic against an independently initialized model (seed 1000). Figure 2 (Upper Right) shows that the correlations for same-initialization models, though small in absolute terms, are consistently shifted upward relative to the independent reference and stabilize rather than decaying to zero. Initialization thus leaves a weak but statistically detectable bias signal that survives training—the persistence property on which our fingerprint rests.

4 ALGORITHM

Section 3 establishes that models sharing an initialization lineage retain weak but consistent agreement on a subset of output dimensions—those most strongly disfavored by the untrained model under random inputs. This motivates a concrete algorithmic objective: given any two models, and *without* access to the underlying initialized model, recover a set of dimensions that carry these bias signals and determine, through a calibrated statistical test, whether the two models’ agreement on them is significant. The procedure comprises three stages—localizing the biased dimensions, forming a correlation distribution over them, and testing that distribution against a null—summarized in Algorithm 1 and detailed component by component below.

Extracting identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . Each input x_i admits two interchangeable instantiations: uniform sampling of ℓ token IDs from the vocabulary, or direct sampling of ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality and $g(x_i)$ is taken as either the model’s logits or its final-layer representations. We then identify the model’s disfavored dimensions as the bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We localize the bias signal through the mean response vector rather than the per-sample argmin frequency (the count of how often each dimension attains the minimum value, as illustrated in Figure 2) because the mean estimator is provably more stable and noise-robust, and also performs better empirically; Appendix C.1 makes this comparison precise.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. Two models sharing an initialization lineage independently recover overlapping sets of disfavored coordinates, so their bottom- m sets exhibit non-trivial intersection. The intersection criterion thus operates as a form of mutual verification: it suppresses spurious agreement arising from architectural similarity or shared training dynamics and isolates the bias signals most plausibly attributable to a common initialization. When one model is closer to initialization, its disfavored dimensions can additionally serve directly as an anchor.

Correlation statistics. Restricting both models to the identity dimensions $\mathcal{S}$, we compare their responses across the random inputs.¹ For each $j \in \mathcal{S}$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This yields a set of correlation statistics $T = \{\tau_j\}_{j \in \mathcal{S}}$ characterizing the distribution of agreement between the two models across identity dimensions. Consistent with Section 3, same-lineage models exhibit distributionally higher agreement on these dimensions than unrelated models do.

Hypothesis testing against a null distribution. Because the correlation distributions of same-lineage and different-lineage models can overlap, thresholding individual statistics is inadequate; we instead test whether the observed correlations are significantly larger than expected under the null hypothesis of no lineage relation. The ideal null is the correlation distribution between independent models, as in Figure 2, but estimating it empirically demands comparisons across many independently initialized models and is therefore computationally expensive. Because this null is well approximated by a Gaussian (Appendix B), a substantially cheaper simulation-based alternative becomes available: an empirical null constructed by passing Gaussian surrogates through the identical pipeline, as summarized in Algorithm 1.

Although this simulation-based null already yields a strong empirical approximation, it incurs avoidable randomness and computational overhead: its precision is bounded by the number of simulation trials, its outcome varies across runs, and comparing two empirical distributions imposes further assumptions from the chosen test (e.g., the one-sided t -test or the Mann–Whitney U test). Since the null distribution of Kendall–Tau is available in closed form under weak independence, we

¹Prior to computing correlations, we apply a row-wise softmax normalization to each sample’s output vector. This eliminates scale differences across models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau; a relatively high temperature avoids degenerate one-hot behavior.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{S}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

take a further step and replace the simulation with an analytical test, retaining both the empirical and the analytical implementations in our code repository. Concretely, under the null hypothesis each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ grows, the distribution of the aggregate statistic $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. Softmax normalization introduces only weak cross-dimensional dependencies, which the relatively high temperature ($T = 10$) further attenuates, so that the independence assumption holds as an accurate approximation in practice; we verify empirically that this Gaussian approximation closely matches the distribution obtained from the full pipeline (Appendix B). We therefore compute the one-sided z -score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$ and the

associated p -value $p = 1 - \Phi(z)$, with Φ the standard normal cumulative distribution function, and reject the null—declaring shared lineage—whenever $p < 0.01$. The outcome is a direct, calibrated statistical decision rather than a heuristic similarity comparison.

Practical instantiation. Model outputs may be taken as either logits or hidden representations; both are consistent with our formulation. Logits, however, require substantially more random inputs to estimate bias patterns reliably as the vocabulary grows, incurring considerable computational overhead (Appendix C.2 provides the supporting sample-complexity analysis). For efficiency and stability, we therefore adopt final-layer hidden states as the default output. The resulting p -value constitutes a direct statistical test for lineage that dispenses entirely with heuristic similarity thresholds.

5 EXPERIMENTS

Our evaluation is organized into two parts that together substantiate the paper’s central claim. Section 5.1 establishes that SeedPrints functions as a genuine fingerprint along two axes: (i) it enables *birth verification at the seed level*, and (ii) it remains *verifiable across the entire training lifecycle*. Existing baselines satisfy neither property—they cannot support verification during early pretraining and, relying chiefly on data-dependent signals, collapse under substantial distribution shift. Unless otherwise stated, we report results using the analytical null with *hidden-state* outputs; complementary results using the empirical null with both *logit* and *hidden-state* outputs, evaluated under the one-sided t -test and the Mann–Whitney U test, appear in Appendix D.2 and corroborate the same conclusions. Section 5.3 then evaluates all methods under practical infringement scenarios on *LeafBench* (Shao et al., 2025). Across 65 distinct open-weight model instances spanning 6 representative post-development techniques, SeedPrints matches the strongest baselines in post-training settings while remaining robust across the full range of deployment transformations.

Baselines. We compare against four established passive fingerprinting baselines spanning weight- and representation-based signatures. *Intrinsic fingerprint* (Yoon et al., 2025) (elsewhere termed *PDF*) compares models through the similarity of the layerwise standard-deviation profiles of their attention parameters. *REEF* (Zhang et al., 2024) measures centered-kernel-alignment (CKA) similarity between the feature representations that two models produce on identical samples. *PCS* and *ICS* (Zeng et al., 2024) (collectively *HuRef*) are weight-similarity methods: PCS flattens all parameters and computes cosine similarity, whereas ICS first constructs invariant terms from the weights and then

Table 1: SeedPrints separates models “at birth”: for every pair of differently seeded initializations the p -value exceeds 0.01, so shared lineage is (correctly) never asserted.

Model Pairs	p -value (> 0.01)
s_{42} VS. s_{2000}	0.116
s_{123} VS. s_{42}	1.000
s_{1000} VS. s_{123}	0.327
s_{2000} VS. s_{1000}	1.000

Table 2: Training preserves the initialization fingerprint. Each trained model s^{base} is matched to its own initialization s^{init} : SeedPrints returns $p < 0.01$ for every pair (✓), whereas all four baselines fall below their 0.8 threshold and fail (✗)—direct evidence that their separability rests on training-induced rather than initialization-born signals.

Model Pairs	SeedPrints p -value	Baselines			
		Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.37e-26 ✓	-0.021 ✗	0.375 ✗	0.580 ✗	0.196 ✗
s_{123}^{init} vs. s_{123}^{base}	3.16e-33 ✓	0.149 ✗	0.369 ✗	0.581 ✗	0.188 ✗
s_{1000}^{init} vs. s_{1000}^{base}	1.12e-21 ✓	-0.252 ✗	0.381 ✗	0.581 ✗	0.188 ✗
s_{2000}^{init} vs. s_{2000}^{base}	3.70e-31 ✓	-0.337 ✗	0.331 ✗	0.581 ✗	0.188 ✗

computes cosine similarity on those invariants. In Section 5.3 we additionally include the weaker gradient-based fingerprint *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), all similarity-based baselines use a **0.8 similarity threshold** for binary decisions.

Throughout the experiment tables, cell color encodes ground-truth lineage: **green** marks pairs drawn from the same source, and **red** marks pairs from different sources. For instance, s_{42}^{init} vs. s_{42}^{base} contrasts a model initialized with seed 42 against its continued-pretrained descendant (same source, hence green), whereas s_{2000}^{init} vs. s_{42}^{base} contrasts a seed-2000 initialization with a model trained from a seed-42 initialization (different sources, hence red). A ✓ denotes a correct detection and a ✗ an error, so that the agreement between symbol and cell color conveys each method’s accuracy at a glance.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) entirely from scratch. The main paper reports LLaMA-style results and defers the Qwen-style counterparts to Appendix D; the conclusions coincide across both families, indicating that the phenomena we report are not specific to a single architecture.

Different initialization seeds produce distinct fingerprints. Table 1 reports the p -values of our correlation test for pairs of models initialized with different random seeds (42, 123, 1000, and 2000). Every p -value exceeds the 0.01 threshold, so the test correctly declines to attribute shared lineage in all cases: distinct seeds yield distinct fingerprint behaviors, and SeedPrints separates models “at birth.”

Training preserves the initialization fingerprint. Table 2 pairs each initialization model s^{init} with its descendant s^{base} , obtained by pretraining on OpenWebText (Gokaslan et al., 2019) (≈ 10 B tokens). Across every seed–model pair, SeedPrints returns p -values far below 0.01 (as low as 3.16e–33), certifying that the bias profiles remain strongly correlated and that each descendant inherits its initialization fingerprint intact. Evaluated on the identical pairs, all four baselines fail without exception to link an initialization to its own descendant—direct evidence that their separability derives from training-induced artifacts rather than from the initialization itself.

Identical data and training order do not make fingerprints converge. In Table 3, the four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are all trained on *exactly the same corpus* (OpenWebText) in *exactly the same data order*—we fix the training seed to lock the curriculum—so that their sole point of difference is the initialization seed i . Across every cross-seed pair the p -values remain above 0.01, in stark contrast to the near-zero values of Table 2. The fingerprint is therefore determined by initialization alone: it stays seed-specific even when data and curriculum are held identical, isolating a signal that no shared training pipeline can manufacture.

Continual training on divergent datasets does not confound the fingerprint. The experiments above establish SeedPrints as a seed-level biometric fingerprint. From a copyright standpoint, the inability of prior methods to separate seeds is not in itself a liability, since an initialization seed carries no clear copyright status; their genuine fragility lies elsewhere. Their attribution is easily misled by *data distribution, failing to recognize lineage once the training distribution shifts substantially during continued training.*

Table 3: Identical data and training order do not make fingerprints converge: models sharing corpus and curriculum but differing in initialization seed still yield $p > 0.01$, confirming that the fingerprint is fixed by initialization alone.

Model Pair	p -value (> 0.01)
s_{42}^{init} vs. s_{123}^{base}	0.573
s_{123}^{init} vs. s_{2000}^{base}	0.724
s_{2000}^{init} vs. s_{1000}^{base}	0.109
s_{1000}^{init} vs. s_{42}^{base}	0.883

Table 4: Fingerprint persistence under continual training on divergent corpora (base model: seed 1000, pretrained on openwebtext). Green rows are true descendants and red rows are distractors. SeedPrints attributes lineage correctly in every case, including the sharply out-of-distribution `the_stack`, whereas the baselines track domain similarity and misclassify the true code descendant.

Setting	SeedPrints	Baselines			
Continual corpus (seed)	p -value	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	≈ 0 ✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	≈ 0 ✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	1.000✓	0.445✓	0.580✓	0.301✓	0.026✓

To expose this failure directly, in Table 4 we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two deliberately dissimilar corpora: TinyStories (Eldan & Li, 2023), comprising synthetic children’s stories, and The Stack (Kocetkov et al., 2022), comprising permissively licensed GitHub code. We contrast (i) true descendants continued from the base against (ii) *distractors* derived from a different base model (seed 123). Because the baselines are insensitive to initialization, the distractor base is pretrained with a different data order on OpenWebText and then continued on the *same* target corpus; the task is to determine which descendant truly shares lineage with the base.

Under the code setting (The Stack), every baseline fails, misclassifying the true descendant as a distractor. This confirms that the baselines track domain similarity rather than lineage identity: TinyStories remains close in distribution to the OpenWebText pretraining corpus, whereas The Stack diverges sharply, and this pronounced distribution shift alone suffices to defeat detection. SeedPrints, by contrast, attributes lineage correctly on both corpora. The fingerprint is therefore no proxy for data distribution—it withstands pronounced domain shift and persists well beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. Does SeedPrints reliably recognize its own offspring along an entire training trajectory? We answer this first in a controlled pretraining experiment on OpenWebText, where the initialization seed is known: testing lineage at intermediate checkpoints yields consistent detection at every stage (Appendix D.2.1, Figure 5).

Large-scale pretraining. To probe lineage under realistic foundation-model training, we analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens), treating the final checkpoint as the target model and each earlier checkpoint as a candidate ancestor. Unlike the controlled OpenWebText setting, this trajectory spans heterogeneous corpora, far longer optimization, and large-scale representation drift. For directional consistency with the similarity-based baselines we plot $1 - p$; across all checkpoints the p -values are $\ll 0.01$ and numerically indistinguishable from zero.

As Figure 1 shows, **every existing baseline fails to verify lineage during early pretraining** (within the first **trillion** tokens, $ICS \approx 0$, $PCS < 0.3$, and $REEF < 0.6$), whereas SeedPrints remains verifiable throughout and strengthens monotonically as training proceeds. This exposes a blind spot in current evaluation practice: **prior work concentrates on verifying lineage only after fine-tuning, where detection is substantially easier**. Our results show that large-scale pretraining itself markedly reshapes representations and amplifies the lineage signals these baselines depend on, so that their late-stage success can convey a *false sense of safety*. Reliable auditing must therefore engage the early pretraining stages—where misuse is hardest to detect and where, among the methods evaluated, only SeedPrints remains effective—rather than relying on late-stage checkpoints alone.

Large-scale fine-tuning. We next benchmark SeedPrints against the baselines under standard fine-tuning-stage evaluation, testing suspect models fine-tuned from Llama-2-7b across fine-tuning volumes that span five orders of magnitude, from 5 million to 700 billion tokens. The suspects comprise diverse downstream variants—Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang

Table 5: Fingerprinting results under large-scale fine-tuning, each row comparing a target model against its LLaMA-2-7B base across fine-tuning volumes spanning 5M to 700B tokens. **SeedPrints** reports the p -value from our correlation test (< 0.01 signals shared lineage), whereas the four baselines report similarity scores (threshold = 0.8, higher is better). SeedPrints certifies lineage in every setting, while the weight-similarity baselines PCS and ICS fall below threshold on the two most extensive (500B–700B-token) code fine-tunes.

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000✓	0.9950✓	0.9979✓	0.9952✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000✓	0.9985✓	0.9985✓	0.9949✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000✓	0.9979✓	1.0000✓	0.9994✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990✓	0.9978✓	1.0000✓	0.9817✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480✓	0.9947✓	0.6863✗	0.3369✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470✗	0.9984✓	0.6682✗	0.2905✗

et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), CodeLlama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023)—trained on 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As Table 5 reports, SeedPrints maintains $p < 0.01$ in every case, whereas the weight-similarity baselines PCS and ICS drop below their detection threshold under the two heaviest (500B–700B-token) code-oriented fine-tunes—confirming that the initialization fingerprint remains legible even after adaptation far more extensive than typical downstream use.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployment routinely subjects foundation models to *parameter-altering* adaptations—fine-tuning, PEFT, quantization, and beyond—each of which can weaken or distort a fingerprint signal. SeedPrints is designed to remain effective throughout this regime, and we test that claim on *LeaF-Bench* (Shao et al., 2025), a dedicated benchmark for language-model copyright auditing built from open-weight models.

Across source models. We audit two types of source model, *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each, the task is to separate *derivative* models—obtained via post-training adaptation of the source—from *independent* models trained without access to the source weights, mirroring the practical scenario in which a lineage claim must be adjudicated for either a PT base or its IT variant. The evaluation spans 65 models in total; ground-truth lineage follows the derivation relationships documented in each model’s HuggingFace metadata, and the complete roster of model names and repositories appears in Appendix E.

Metrics. We quantify detection performance with (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), the latter measuring the maximum separation between the score distributions of derivative and independent models.

SeedPrints produces a per-model p -value from a statistical test, an object of a different nature from the similarity scores of the baselines: similarity scores are linear quantities whose differences carry proportional meaning, whereas a p -value is a tail probability expressing the rarity of an observation under the null and does not live on a linear scale. To render the two comparable, we map our p -values to scores via $s = 1 - p$. This conversion is, if anything, disadvantageous to our method—small p -values vary on an exponential scale while threshold sweeping is linear, blurring fine-grained distinctions—yet SeedPrints attains comparable or superior performance even under this handicap. Accordingly, we (i) compute AUC for all methods by sweeping thresholds over the scores s and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol preserves compatibility with the baseline pipelines while retaining the statistical interpretability of our test. We omit Manhattan distance, since absolute distances between p -values are not meaningful.

In most practical settings, calibrating an optimal similarity threshold is infeasible, and it is precisely here that our formulation offers a distinctive advantage. Although we report AUC for comparability, SeedPrints differs from prior methods in kind rather than degree: it constitutes a direct statistical test. **For any pair of models it emits a p -value that supports a same-lineage decision without any tunable threshold**, yielding reliable verification under operational conditions in which threshold-based methods cannot be calibrated.

Table 6: Family-wise comparison of LLM fingerprinting methods across source models, where “PT” and “IT” denote pre-trained and instruction-tuned source models. The per-family breakdown covers the 6 families possessing both PT and IT models; “Overall” scores aggregate all test pairs under a single global threshold and are therefore not the average of the per-family scores. SeedPrints, Intrinsic, and ICS are all essentially saturated (overall AUC ≥ 0.992), placing SeedPrints on par with the strongest dedicated post-training baselines, whereas REEF and Gradient trail substantially.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC $\uparrow$	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic $\uparrow$	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC $\uparrow$	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic $\uparrow$	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC $\uparrow$	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC $\uparrow$	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic $\uparrow$	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

Parameter-altering techniques. To stress-test robustness, we evaluate models produced by six representative transformations: (1) instruction tuning (Instruct), (2) general-purpose fine-tuning (Finetune), (3) parameter-efficient fine-tuning (PEFT), (4) quantization, (5) model merging (Merge), and (6) distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

The pool comprises seven model families. For each family we treat both its pretrained base (PT) and its instruction-tuned variant (IT) as separate source models and compare each against all other models in the pool of 65, with results reported in Table 6. We exclude the TinyLlama-1.1B family from Table 6, as it constitutes a trivial case for every method. Each column corresponds to one family, whose AUC is computed over the associated tests using a family-calibrated threshold, whereas the overall score aggregates all test pairs under a single global threshold; the overall score is consequently not the average of the per-family scores. Three methods—*Intrinsic*, *ICS*, and our *SeedPrints*—are essentially saturated across all alterations, with overall AUC of 0.994, 0.994, and 0.992 and KS of 0.989, 0.943, and 0.986, respectively, and attain the maximal score in most families. The remaining baselines, *REEF* and *Gradient*, trail substantially. SeedPrints thus delivers reliable, persistent identity tracking across these diverse open-weight families: although designed to capture the intrinsic seed-level fingerprint, it simultaneously matches the strongest post-training baselines and functions as a full biometric-grade identifier.

5.3.2 ALTERATION-WISE ROBUSTNESS

Disaggregating by alteration type, *Instruct* is uniformly the easiest case—even the weakest baseline, *Gradient*, reaches AUC 0.895—whereas *Finetune* and *Merge* are the most demanding, no method attaining perfect performance on *Finetune*. The remaining transformations elicit sharply method-dependent behavior: REEF degrades markedly under *Quantization* (0.996 $\rightarrow$ 0.871) and *Distillation* (0.996 $\rightarrow$ 0.858) yet stays resilient to *PEFT* (0.994), while *Gradient* instead struggles with *PEFT* and *Quantization* (0.895 $\rightarrow$ 0.776/0.756). The three stronger methods remain robust throughout, exceeding 0.9 on every transformation. SeedPrints dips only to 0.959 on *Merge*, a dip traceable to a single misclassified merge model out of seventeen; this lone error indicates that weight-space interpolation with unrelated parameters can occasionally dilute the initialization signal below the detection threshold, and *Merge* is the sole transformation on which SeedPrints is not essentially saturated. Averaged over all six alteration types, *Intrinsic* holds the highest overall AUC, with *ICS* and *SeedPrints* following closely—so that SeedPrints matches the best dedicated post-training fingerprints while additionally covering the pretraining regime in which they fail.

5.4 ABLATION STUDY ON HYPERPARAMETERS

SeedPrints exposes three hyperparameters: the number of random input sequences n , the length of each random sequence l , and the fingerprint size m . The first two jointly govern the amount

Table 7: Robustness of LLM fingerprinting methods across six parameter-altering transformations. SeedPrints stays at or near saturation on five of the six—dipping only to 0.959 AUC on *Merge*, attributable to a single misclassified model—thereby matching the strongest baselines, whereas REEF and Gradient degrade markedly under quantization, distillation, or PEFT.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

of random input, while m fixes how many low-mean coordinates enter the test. Throughout the main experiments in Sections 5.2 and 5.3 we set $n=2000$, $l=1024$, and $m=0.1 d_{\text{out}}$, where d_{out} is the hidden-state dimensionality, so that the configuration adapts automatically to open-weight models of differing width. To characterize sensitivity, we vary each hyperparameter in turn while holding the others at their defaults ($n = 2000$, $m = 400$, $l = 768$) and report empirical accuracy and false-positive rate at a fixed significance level $\alpha = 0.01$ in Table 8.

Table 8: Ablation study under significance level $\alpha = 0.01$.

Increasing either n or l consistently improves identification, a trend our theory in Appendix C.2 predicts precisely: accurate recovery of the population-level bias requires the query count to satisfy $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$, where γ is the bias margin separating the biased coordinates from the rest. Because each target dimension demands a minimum sample budget for reliable detection, an insufficient n necessarily degrades performance. The margin γ itself grows with l through accumulation effects, so longer sequences sharpen the separation; under a fixed and otherwise insufficient number of sequences, increasing l therefore lowers the effective query requirement and restores detection accuracy.

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

The effect of m is non-monotonic, reflecting a bias–variance trade-off. Too small an m leaves the selected coordinates unstable, as too few empirical observations remain to form a reliable distribution; too large an m admits many irrelevant coordinates whose noise shrinks γ and, by the sample-complexity bound above, would demand quadratically more samples to offset. Empirically, m in the range 200–400 performs well across model sizes, consistent with our heuristic $m \approx 0.1 d_{\text{out}}$ for the typical range $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

We introduced *SeedPrints*, a stronger and intrinsic notion of LLM fingerprinting that traces a model’s lineage back to its random initialization. Our central finding is that initialization itself imprints a persistent, seed-specific fingerprint on neural language models: untrained models already exhibit stable, seed-dependent token-preference patterns, and these biases remain statistically measurable throughout training. Building on this finding, we developed a distribution-correlation test over *identity dimensions* that certifies shared lineage relative to a candidate reference at a calibrated significance level, replacing heuristic similarity thresholds with a direct hypothesis test. Across from-scratch LLaMA- and Qwen-style families and a broad suite of realistic deployments—continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation—SeedPrints delivers “birth-to-lifecycle” verification, succeeding in the early-pretraining regime where prior methods fail while matching the strongest baselines afterward. Within its white-box scope it thus constitutes a practical and dependable tool for provenance and copyright auditing.

Limitations and Future Directions Our method reads out fingerprint signals rooted in initialization-stage bias, and it operates within the white-box paradigm: it requires access to model outputs—logits or final-layer hidden states—over many random inputs, and therefore does not directly apply under restricted, top- k , or text-only API access. It likewise verifies lineage relative to available reference models or checkpoints rather than recovering a numeric seed in isolation. We regard these as boundaries of the present instantiation rather than of the underlying principle. Indeed, the pursuit of *true fingerprints* for LLMs remains at an early stage—most existing approaches, black-box methods especially, depend on signals that surface only in later training or from alignment-driven artifacts and cannot robustly protect models at the pretrained stage—which makes reliable all-stage fingerprints all the more pressing. By demonstrating that such a signal exists and is recoverable from initialization onward, our work marks a concrete step toward that goal; the full discussion appears in Appendix [A.2](#).

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A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We employ AI assistants—specifically ChatGPT and Gemini—for writing and formatting support, confined to grammar and style checking, improving the clarity of figure and table captions, and comparable surface-level edits. For programming tasks, we occasionally use GitHub Copilot and Claude as coding assistants, for example for code auto-completion and debugging hints. In all cases the underlying scientific content, experiments, and analysis are our own.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

While this work establishes seed-born, training-persistent biases as reliable fingerprint signals, several boundaries delimit the present instantiation and mark directions for future work.

Access requirements. SeedPrints operates within the white-box paradigm: it estimates output-preference bias from a model’s logits or final-layer hidden states over many random inputs, and therefore does not directly apply when only restricted API access is available—for instance, when a service exposes only top- k logits, sampled text, or deterministic and safety-filtered responses. Extending the signal to such black-box conditions, where a substantial performance gap relative to white-box methods remains, is a challenging open direction that will require significant further work.

Reference-relative verification. The test adjudicates whether a suspect model shares an initialization lineage with a supplied reference model or checkpoint; it does not recover a numeric seed in the abstract. In practice, the candidate references are the very models or checkpoints whose ownership is disputed, so this reference-relative formulation matches the operational structure of provenance auditing.

Scope of evaluation. Our study concerns decoder-only, open-weight language models under standard initialization schemes, and its robustness has been demonstrated across a broad range of parameter-altering transformations. Behavior under more aggressive structural interventions—vocabulary resizing, tokenizer replacement, heavy pruning or sparsification, layer replacement, or nonstandard initialization distributions—lies outside the settings we evaluate and is a natural target for subsequent study.

We emphasize that the pursuit of *true fingerprints* for LLMs is still at an early stage even under the white-box setting: most existing approaches rest on training-induced signatures or alignment behaviors and consequently cannot robustly protect pretrained or early-stage models. By analogy with human fingerprints, which remain stable irrespective of age or environment, a true LLM fingerprint should persist across continued training and deployment contexts rather than encode incidental training artifacts—precisely the property SeedPrints attains within its scope. We hope this work motivates continued research toward principled, robust fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The hypothesis test of Section 4 adjudicates shared lineage by comparing the distribution of two models’ correlation statistics over random inputs against a null distribution—the distribution of correlation statistics between two *independent models evaluated on random inputs*.

We approximate this null in two complementary ways: (i) an empirical, simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. This section justifies both approximations in three steps: we characterize the true null using independently initialized models, introduce the Gaussian surrogate as a tractable simulation-based proxy, and validate the analytical approximation against it.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null is the distribution of correlation statistics between two independent models evaluated on random inputs, and the cleanest realization of this independence is a pair of independently initialized

models—distinct random seeds, no training. We estimate this empirical null over ~ 2500 such pairs, computing Kendall–Tau correlations on 10,000 random inputs exactly as in Section 3. Figure 3 confirms that the resulting true null is empirically Gaussian and centered at zero, so that no pair of independently initialized models systematically agrees on the identity dimensions—the property a well-calibrated null must satisfy.

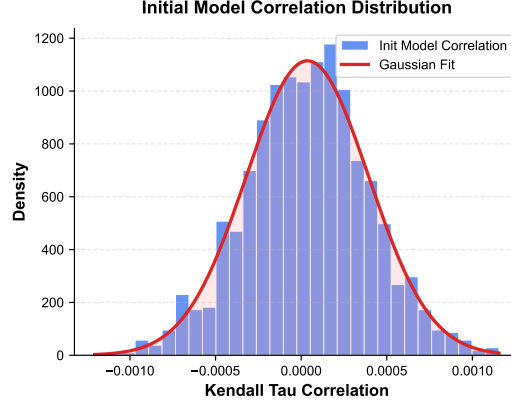


Figure 3: The empirical correlation distribution over ~ 2500 pairs of independently initialized models, evaluated on random inputs, closely matches a Gaussian centered at zero—validating the null model that underlies our hypothesis test and the false-positive control it affords.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Estimating the null directly from independently initialized models is computationally expensive, requiring access to many models and repeated evaluations. We therefore introduce a tractable simulation-based null that substitutes Gaussian surrogates for model outputs under the null.

Two considerations motivate this choice. First, approximating neural-network outputs by Gaussians is well established in the auditing literature, supported by both empirical and theoretical work (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends solely on relative ordering and selection operations—rank correlations and bottom- m selection—and never invokes semantic structure in the inputs; combined with the fact that randomly initialized models have independent parameter matrices and hence exhibit no strong structured output dependencies under the null, this renders independent Gaussian matrices a faithful and tractable proxy for the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and passing them through the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

Because this construction applies the identical selection procedure to Gaussian samples, it automatically captures the dependency patterns introduced by the pipeline itself, such as those arising from selecting $\mathcal{S}$ across dimensions. As Figure 4 (middle) shows, the resulting distribution reproduces the centered Gaussian shape of the empirical null obtained from independently initialized models, establishing the Gaussian surrogate as an effective and inexpensive simulation-based approximation of the true null.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate faithfully reproduces the null under the full pipeline, it still requires repeated sampling and injects randomness into the estimate. We therefore derive a fully analytical—and deterministic—approximation.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . Were the correlations across identity dimensions independent, the central limit theorem would give

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization does introduce cross-dimensional dependencies. Nonetheless, Figure 4 shows that the analytical Gaussian closely matches the distribution produced by the Gaussian surrogate pipeline across settings, spanning varying numbers of random inputs n and output dimensions D . These dependencies are therefore sufficiently weak that they do not materially degrade the approximation—plausibly because the relatively high temperature ($T = 10$) smooths the output distribution and attenuates coupling between dimensions.

Taken together, these results justify the analytical Gaussian null of Section 4 as an efficient, deterministic, and accurate approximation that obviates costly empirical baseline construction while preserving calibrated false-positive control.

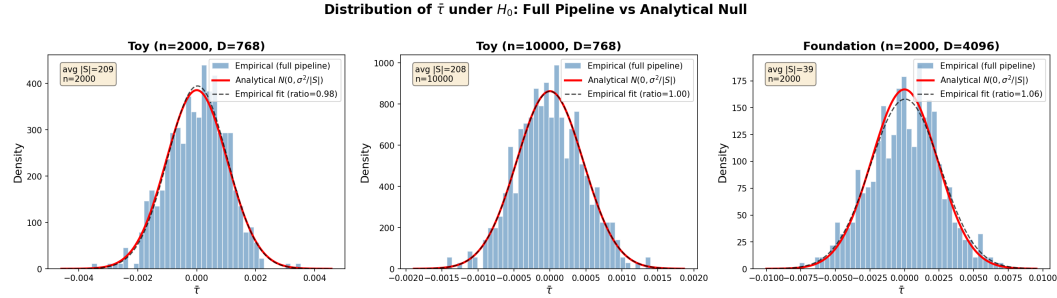


Figure 4: Validation of the analytical Gaussian null. Applying the full pipeline—identity-dimension selection and correlation aggregation—to Gaussian surrogate outputs yields a distribution of $\bar{\tau}$ that closely matches the analytical Gaussian $\mathcal{N}(0, \sigma^2/|\mathcal{S}|)$ across the settings shown. The agreement demonstrates that the analytical approximation captures the null behavior accurately despite the weak dependencies the pipeline introduces.

On variance scaling. The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation, but this mismatch leaves our hypothesis-test conclusions intact, and it does so in the conservative direction. The primary simulation-based evaluation in Appendix D.2 uses the Mann–Whitney U -test, which is rank-based and hence invariant to any positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$, so all rank statistics and rejection thresholds are unchanged. For variance-sensitive tests, we additionally evaluate the t -test, whose statistic scales as $t' = t/c$ when the variance decreases. Since $c < 1$ here, the scaling shrinks $|t|$ and renders the test *more* conservative rather than inflating significance. Empirically, across the extensive experiments of Appendix D.2, this variance scaling leaves the power of the t -test unaffected, indicating that the separation between the alternative and the null far exceeds the scaled variance.

For the analytical null used in the main paper, the variance enters explicitly in the z -score normalization, so any scale discrepancy is absorbed into the denominator. Because the analytical variance is slightly larger than the empirical one, the resulting z -scores are smaller, again rendering the test conservative and guarding against false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

Our objective is to identify the output dimensions that an initialized model is least likely to predict under random inputs. These dimensions correspond to directions the model inherently disfavors owing to initialization-induced bias—independently of the input distribution—and thereby form the

²As discussed in Appendix B.2, the outputs may exhibit weak dependencies induced by shared inputs, but this approximation remains empirically accurate.

basis of a stable fingerprint signal. This section formalizes that objective and establishes why the mean-based extractor we adopt recovers it stably.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings, as in the main paper). Each output coordinate $g_j(x)$, for $j \in [d_{\text{out}}]$, is then a random variable with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we seek to capture are the m output dimensions of smallest expected value, formally

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap quantifies how well-separated the target set $\mathcal{M}$ is from the remaining coordinates. Crucially, for a fixed m and input distribution, $\Gamma(m)$ is governed solely by the model’s inherent initialization bias. *Moreover, lengthening the random input sequences amplifies this bias through concentration, sharpening the separation at the population level.*

Two natural empirical strategies present themselves for extracting these population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by its empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates of smallest value. Following the notation of Section 4, in which g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select its smallest-value coordinate and then retain the m most frequently selected dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both strategies are intuitively reasonable, yet they differ sharply in stability. As we prove below, sample-mean-based extraction is provably stable, whereas per-sample argmin voting applies a discontinuous operator that can flip under arbitrarily small perturbations, rendering it inherently noise-sensitive—which is exactly why we adopt the former.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first establish that the sample-mean-based operator, which selects the m smallest coordinates of the empirical mean vector, is stable under small perturbations to model outputs—perturbations that model realistic variation in the outputs of the *same* model, such as numerical noise or evaluation under differing precision modes. This stability is the property that lets identity dimensions be recovered consistently across the two models being compared.

Formally, we write perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ denotes the stochastic deviation. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\hat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma furnishes a quantitative upper bound on the probability that the sample-mean-based extractor alters its selected set under perturbation. This instability probability decays exponentially in the number of averaged samples n , a direct consequence of variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\hat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

For any coordinate j , by definition of $\hat{g}'$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \right\}.$$

Equivalently, if $\max_j |\hat{g}'_j - \hat{g}_j| < \hat{\Gamma}(m)/2$, then the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\hat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \hat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \hat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \hat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\hat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\hat{\Gamma}(m)}{2}) \geq \hat{\Gamma}(m) - \hat{\Gamma}(m) = 0.$$

Thus no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$ under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\hat{\Gamma}(m)$ is fixed under conditioning on X , plugging $\varepsilon = \hat{\Gamma}(m)/2$ into (2) yields

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special case of this assumption.

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

By contrast, per-sample voting applies a discontinuous winner-take-all operator to each input, so that arbitrarily small output perturbations can flip its decisions with non-vanishing probability. We retain the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$ as above.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*}(x; g)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting reaps no benefit from variance reduction: each input triggers a hard decision on noisy outputs, so increasing n merely multiplies the opportunities for index flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) | x)$, and $\alpha := \mathbb{E}_x \left[\Phi \left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma} \right) \right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i | x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E} \left[\prod_{i=1}^n (1 - q(x_i)) \right] \leq \mathbb{E} \left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma))) \right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$ (i.e., only a constant number of votes), then a change in $j^*(x_i)$ for a single input x_i is sufficient to swap the ordering across this boundary and thus alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now determine how many random inputs the empirical mean estimator requires in order to recover the true bias, thereby grounding the hyperparameter trends observed empirically. Reusing the notation of **Setup**, each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$ with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$; we write $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ for the means sorted in non-decreasing order and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ for the separation margin. We estimate $\bar{g}$ from n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimension set is $\mathcal{M}$ of size m , and our empirical estimate is denoted $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We now translate this recovery condition into hyperparameter terms, again adopting the sub-Gaussian assumption of Appendix C.1. The assumption is mild: by Hoeffding's inequality any bounded random variable is σ -sub-Gaussian with σ proportional to its range, and in our setting each $g_j(x)$ is a logit or final hidden activation, which is typically bounded or light-tailed in both initialized networks (owing to controlled parameter initialization) and trained networks alike.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr(\|\hat{g} - \bar{g}\|_\infty > \varepsilon) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\hat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Accurate recovery therefore requires $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$: the number of random queries grows only logarithmically in the output dimension but inversely in the squared separation margin γ . This favorable, near-logarithmic dependence on d_{out} is what keeps the query budget tractable even for large-vocabulary or wide models. The margin γ is in turn shaped by two factors:

- **Intrinsic bias strength.** Longer random input sequences amplify the initialization-induced bias through concentration, widening the population separation between coordinates.
- **Choice of m .** Different values of m probe different regions of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ and hence induce different effective margins in practice.

The former is controlled directly through the input length l , whereas the latter is chosen empirically from the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all from-scratch models with the Hugging Face Transformers Trainer (Wolf et al., 2020) and use Accelerate (Gugger et al., 2022) for distributed runs. Every open-weight model is loaded from its official Hugging Face release and used under its original license—Llama models under the Meta Llama Community License and the remaining models under Apache-2.0—while all datasets are obtained through the Hugging Face Datasets library (itself Apache-2.0), with dataset content governed by each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section provides the complementary results for Section 5. We report results using the empirical simulation-based null of Algorithm 1 with both *logit* and *hidden-state* outputs, evaluated under both the one-sided t -test (t -test) and the Mann–Whitney U test (U test). Across every configuration the conclusions of the main text hold unchanged, confirming that our findings are robust to the choice of null construction, output representation, and test statistic rather than artifacts of any one of them.

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based null is stochastic, we report p -values averaged over 10 independent trials at a significance level of $\alpha = 0.01$. We stress that the absolute magnitude of extremely small p -values carries no interpretive weight: once p drops below the floor set by numerical and sampling noise (e.g., $< 10^{-20}$), a value such as 10^{-260}

Table 9: Seed-level separability under different random initializations across model families: for both LLaMA- and Qwen-style models, and under both the t -test and the U -test, every cross-seed p -value exceeds 0.01, so distinct seeds are never conflated—evidence of controlled false positives across architectures and test statistics.

(a) LLaMA-style models					(b) Qwen-style models				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test		t -test	U -test	t -test	U -test
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style models) Training preserves the initialization fingerprint. SeedPrints returns $p < 0.01$ under both output types and both test statistics for every s^{init} – s^{base} pair, whereas all four baselines fail—corroborating that their separability is training-induced rather than initialization-born.

Model Pair	Logits Output		Hidden State		Baselines			
	t -test	u -test	t -test	u -test	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

constitutes no stronger evidence than 10^{-20} —both reject the null decisively, and only the rejection, not the numeral, should be read.

Different initialization seeds produce distinct fingerprints. Table 9a (LLaMA-style) and Table 9b (Qwen-style) report the p -values of our correlation test for pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under both the t -test and the U -test. Every p -value exceeds 0.01, so the test consistently declines to assert shared lineage between distinct seeds: distinct seeds yield distinct fingerprint behaviors, and models remain separable “at birth” under both test statistics and both architectures.

Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) pair each initialization model s^{init} with its descendant s^{base} trained on OpenWebText (Gokaslan et al., 2019) (≈ 10 B tokens). Across every seed–model pair the p -values fall below 0.01, certifying that the bias profiles remain strongly correlated and share a common lineage: each trained model inherits its initialization fingerprint intact, and training does not erase it. On the same pairs (Table 10) the baselines fail without exception to link an initialization to its descendant, again indicating that their separability originates in training-induced artifacts rather than in the initialization itself.

Identical data and order do not make fingerprints converge. In Table 13a (LLaMA-style) and Table 13b (Qwen-style), the four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) in exactly the same data order*—we fix the training seed to lock the curriculum—so that they differ only in their initialization seed i . The question is whether identical data and order can erase or confound the fingerprint. They cannot: across all cross-seed pairs the p -values remain above 0.01, in sharp contrast to the near-zero values of Table 10. Fingerprints thus remain seed-specific even under an identical corpus and curriculum.

Continual training on diverse datasets does not confound the fingerprint. The experiments above establish SeedPrints as a seed-level biometric fingerprint. From a copyright standpoint, the baselines’ inability to separate seeds is not in itself a weakness, since an initialization seed carries no clear copyright status; their real fragility lies in being easily misled by *data distribution*, *failing to recognize lineage once the training distribution shifts substantially during continued training*.

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two deliberately dissimilar datasets: TinyStories (Eldan & Li,

Table 11: (Qwen-style models) Training preserves the initialization fingerprint: SeedPrints returns $p < 0.01$ for every $s^{init}_{-}s^{base}$ pair under both output types and both test statistics, mirroring the LLaMA-style result.

Model Pair	Logits Output		Hidden State	
	<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>
s^{init}_{123} vs. s^{base}_{123}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s^{init}_{1000} vs. s^{base}_{1000}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s^{init}_{42} vs. s^{base}_{42}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s^{init}_{2000} vs. s^{base}_{2000}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: Fingerprint persistence under continual training on divergent corpora (base model: seed 1000, pretrained on openwebtext); the *U*-test denotes the Mann–Whitney *U* test. Across logit and hidden-state outputs and both test statistics, SeedPrints correctly attributes lineage on both TinyStories and the out-of-distribution the_stack, whereas the baselines are misled by the code-domain shift—corroborating the main-text conclusion under every null-output-statistic combination.

(a) LLaMA-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0✓	0✓	0✓	7.77e-89✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	1.00✓	0.943✓	0.902✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	0✓	1.73e-287✓	0✓	3.09e-69✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	0.616✓	0.479✓	0.732✓	0.831✓	0.445✓	0.580✓	0.301✓	0.026✓

(b) Qwen-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09✓	1.92e-07✓	8.49e-214✓	5.09e-71✓	1.000✓	0.957✓	0.999✓	0.996✓
TinyStories (123)	0.434✓	0.433✓	0.256✓	0.065✓	0.913×	0.199✓	0.328✓	0.039✓
the_stack (1000)	0✓	4.16e-237✓	1.16e-211✓	2.30e-76✓	0.999✓	0.313×	0.995✓	0.976✓
the_stack (123)	0.999✓	0.993✓	0.610✓	0.491✓	0.916×	0.255✓	0.328✓	0.038✓

2023), synthetic children’s stories, and The Stack (Kocetkov et al., 2022), permissively licensed GitHub code. We contrast (i) true descendants continued from the base against (ii) *distractors* derived from a different base model (initialized with seed 123 and pretrained with a different data order on OpenWebText) and then continued on the *same* corpus, asking which descendant an attribution method identifies as truly sharing lineage with the base.

Under the code setting (The Stack), every baseline fails, misclassifying the true descendant as a distractor—clear evidence that they track domain similarity rather than lineage identity: TinyStories stays close in distribution to the OpenWebText pretraining corpus, whereas The Stack diverges sharply, and this large shift alone suffices to bypass detection. SeedPrints, by contrast, attributes lineage correctly on both corpora. Our fingerprint is therefore no proxy for data distribution: it survives pronounced domain shift and persists well beyond the initial pretraining stage.

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

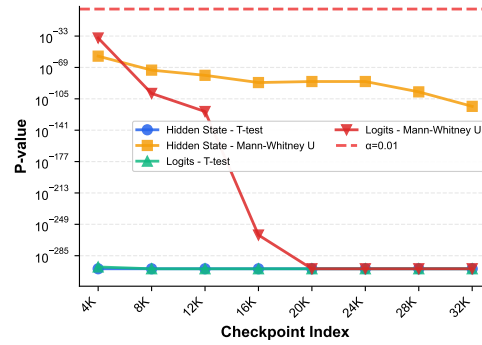
Controlled pretraining trajectory. We conduct a controlled pretraining experiment on OpenWebText with a known initialization seed, treating each intermediate checkpoint as the base and testing whether SeedPrints reliably identifies its offspring along the shared training trajectory. Every variant is consistently recognized as same-lineage, the p -values remaining below the 0.01 threshold throughout (Figure 5a for LLaMA-style models and Figure 5b for Qwen-style models), confirming the stability of SeedPrints under continuous optimization.

Table 13: Identical data and training order do not force fingerprints to converge across initializations: every cross-seed p -value remains > 0.01 under both the t -test and the U -test, for both output types and both architectures.

(a) LLaMA-style models					(b) Qwen-style models				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test		t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: SeedPrints verifies lineage at every checkpoint (p -values < 0.01) for both LLaMA-style and Qwen-style families, evidencing stable all-stage detection under continuous optimization along the controlled training trajectory.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To examine whether SeedPrints “collapses” to a family-level signature as parameter size varies—a natural false-positive concern within a single architecture and tokenizer family—we report cross-size similarity between Qwen-2.5-14B (target) and the smaller models of the same Qwen-2.5 family. Table 14 presents the results for SeedPrint ($\uparrow$) alongside two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$).

Concretely, this cross-size study treats Qwen-2.5-14B as the protected target and considers Qwen-2.5-{0.5B, 1.5B, 3B, 7B} as suspects that share the architectural family yet are not trained as its descendants. As established in the main paper, SeedPrints returns a p -value for the hypothesis “the suspect shares initialization-born fingerprints with the target,” whereas REEF and HUREF are similarity-based baselines that emit continuous scores and require the user to fix a threshold.

Three observations stand out. First, every SeedPrints p -value sits comfortably above a standard significance level (e.g., $\alpha=0.01$), so under our hypothesis-testing view none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This is precisely the desired behavior: despite belonging to the same family and reusing a similar architectural design, these models do not induce a single family-level signature that would spuriously conflate distinct pretraining runs—direct evidence of controlled false positives within a shared architecture and tokenizer.

Second, the cross-size p -values populate different regions of $[0, 1]$ rather than clustering around a single value. We do *not* read these magnitudes as a continuous “similarity score”—their role is to support a calibrated yes/no decision at a fixed significance level—yet their spread nonetheless confirms that the test statistic remains sensitive to the specific combination of architecture and initialization seed rather than degenerating into a generic pattern shared across all Qwen-2.5 variants.

Finally, the similarity-based baselines behave quite differently. REEF yields scores in a narrow band (0.8250–0.8833) across all sizes, while HUREF produces values numerically near zero (on the order of

Table 14: Cross-size comparison between Qwen-2.5-14B (target) and smaller Qwen-2.5 models. SeedPrint reports a p -value ($\uparrow$; every value lies above $\alpha=0.01$, so no smaller model is flagged as lineage-related), whereas REEF ($\downarrow$) and HUREF ($\downarrow$) return threshold-free similarity scores. The fingerprint does not collapse to a family-level signature: it remains sensitive to the specific architecture–seed combination rather than conflating all Qwen-2.5 variants.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

10^{-5}). Lacking any built-in decision rule, these methods force operators to hand-pick thresholds, and it is unclear how a single cutoff could robustly separate cross-size variants from genuinely lineage-related models. SeedPrints, by contrast, delivers a statistical decision at a prescribed significance level, furnishing a clearer and more operational answer to whether two models plausibly share a training lineage within a family.

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers **58** models in total. Below we document the decoding configuration used across all runs and enumerate the model catalog grouped by family, indicating for each derived model its post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, or *distillation*); “PT” and “IT” denote *pre-trained* and *instruction-tuned* source models, respectively. Ground-truth lineage for each derived model follows the derivation relationship declared in its HuggingFace repository.

Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunyncore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemerhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2B-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation applied relative to the PT/IT source model: *finetune* covers supervised and preference-optimization variants; *adapter* covers LoRA/QLoRA-style modules; *merge* covers model soups and TIES-style merges; *quantization* covers GPTQ/INTx variants; and *distillation* covers student models distilled from reasoning-augmented teachers. This taxonomy spans the principal parameter-altering transformations encountered in practical redistribution.

E.2 SAMPLE RESULTS

We present partial results from a single LeafBench evaluation run in Table 15 (*u*-test with the simulation-based null), reporting $1 - p$ as the similarity score. Adopting a significance level of 0.01 for the *p*-values, LeafBench empirically identifies an optimal decision threshold of 0.9920, further attesting to the reliability of the method.

Although individual *p*-values should not be read as linear similarity measures, their distribution across model pairs is highly informative: same-lineage models yield markedly lower *p*-values, whereas unrelated models remain near uniform—exactly the separation a calibrated lineage test should exhibit.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abiliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbiliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample results.** Pairwise similarity ($1 - p$) between target models (rows) and base models (columns). Cells with $1 - p \geq 0.99$ are highlighted in green as same-lineage pairs; the green blocks align cleanly along the true source families, illustrating that same-lineage pairs concentrate near $1 - p = 1$ while unrelated pairs remain diffuse across $[0, 1]$.